Correlation analysis between A. pisum and A. fabae fecundity results

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1 Setup

1.1 Libraries

```
library(dplyr)
library(ggplot2)
```

1.2 Data

```
datapisum_6 <- read.csv( "../../../../02.analyses/01.GWAS/01_phenotipic_analysis/02.ajusted_m

datafabae_4 <- read.csv("../../../../02.analyses/01.GWAS/01_phenotipic_analysis/02.ajusted_m

pisum <- datapisum_6 %>%
  dplyr::select(ivip, emmean) %>%
  dplyr::rename(larvae_pisum = emmean)

fabae <- datafabae_4 %>%
  dplyr::select(ivip, emmean) %>%
  dplyr::rename(larvae_fabae = emmean)

merged <- inner_join(pisum, fabae, by = "ivip")
```

> **Note**
>
> The merged dataset contains 206 IVIPs that were phenotyped for both A. pisum and A. fabae, and for which we have adjusted mean fecundity scores for both aphid species.

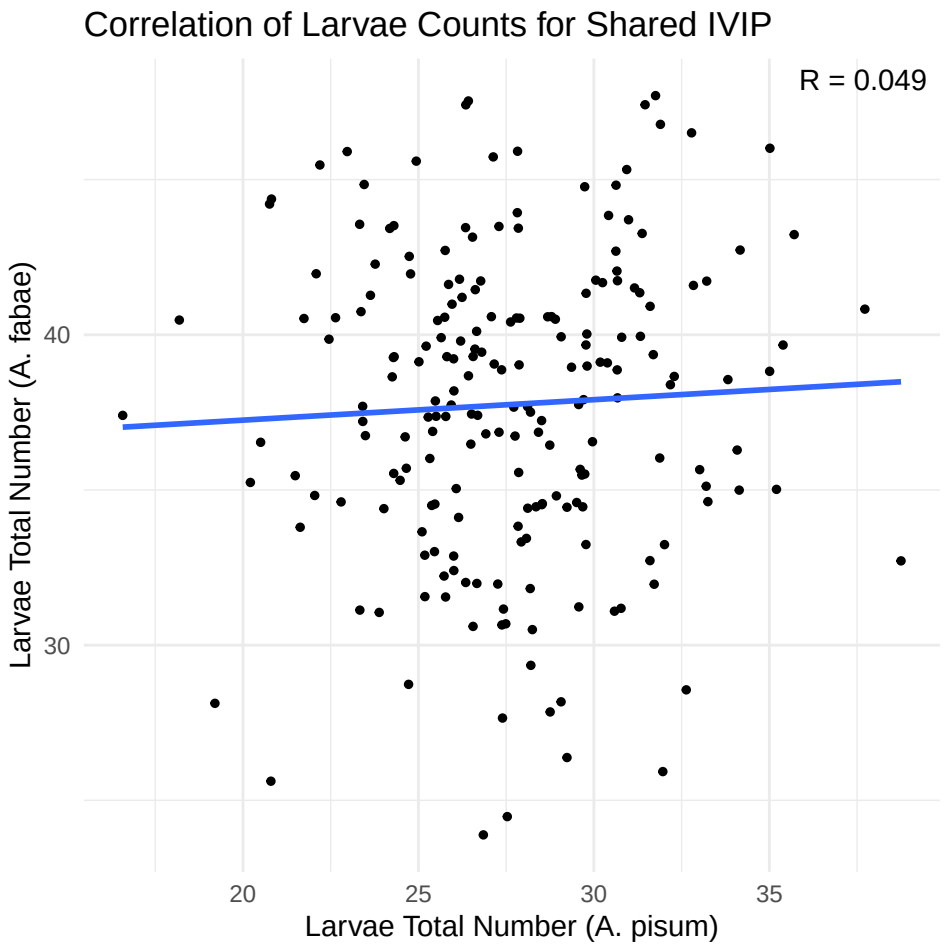
2 Correlation analysis

2.1 Correlation test

```
cor_test <- cor.test(merged$larvae_pisum, merged$larvae_fabae)
```

> **Note**
>
> Correlation between A. pisum and A. fabae fecundity scores for shared IVIPs is weak.
> - R correlation score is equal to 0.049.
> - The corresponding p-value is equal to 0.487 (not significant).

2.2 Scatterplot A. pisum vs A. fabae



3 Genetic clusters and fecundity scores

3.1 Import and add the genetic cluster data to the fecundity data for A. pisum and A. fabae.

```
genetic_cluster <- read.csv("../../../../00.info/02.genetic_structure_jonathan_dijon/faba_bea
  # remove NA records
  filter(!is.na(cluster_name))

merged_cluster <- inner_join(merged, genetic_cluster, by = "ivip")
```

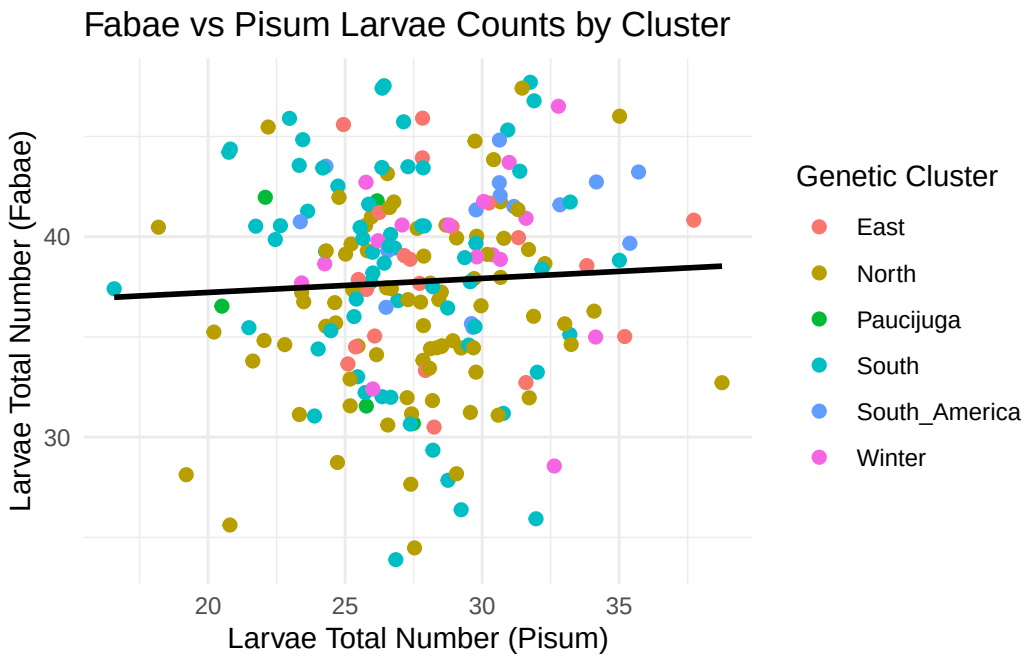
```r
# Ensure cluster_name is a factor
merged_cluster$cluster_name <- as.factor(merged_cluster$cluster_name)



head(merged_cluster)
```

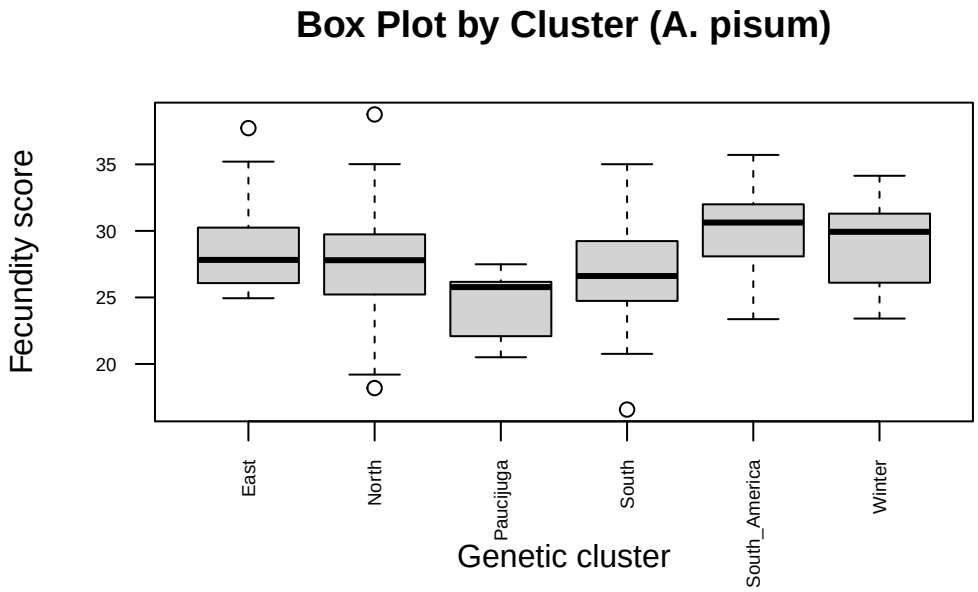
```
        ivip larvae_pisum larvae_fabae cluster_name
1 IVIP100034     26.17123     41.78951    Paucijuga
2 IVIP100081     25.50456     37.37321        North
3 IVIP100121     25.95937     40.98543        North
4 IVIP100123     19.20230     28.12484        North
5 IVIP100124     24.65601     35.69890        North
6 IVIP100159     28.42183     36.85953        North
```

3.2 Scatterplot with genetic cluster info overlaid

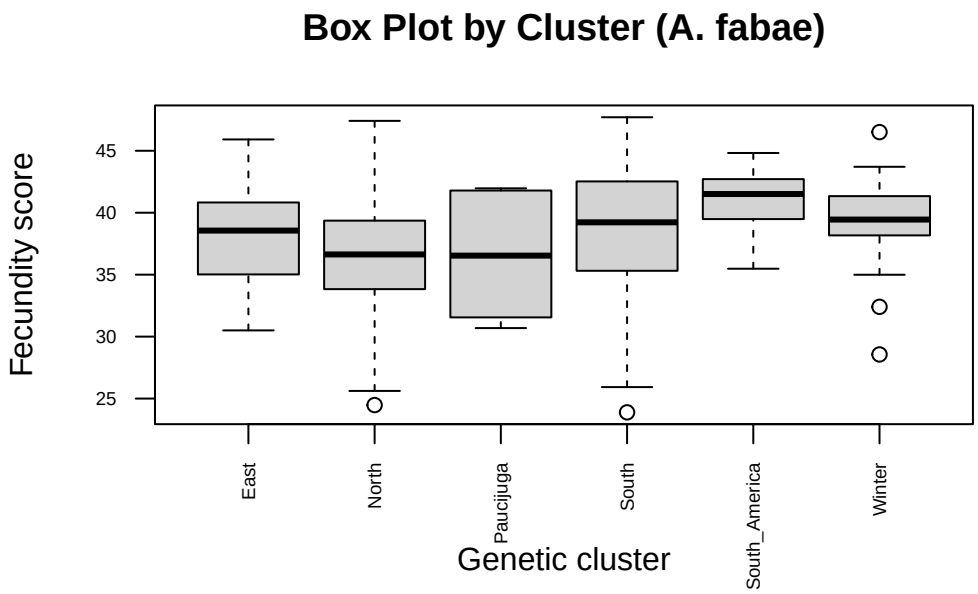


3.3 Boxplot of fecundity scores by genetic cluster for A. pisum and A. fabae, and ANOVA test to check for significant differences between clusters.

```
boxplot(larvae_pisum ~ cluster_name, data = merged_cluster,
        main = "Box Plot by Cluster (A. pisum)",
        xlab = "Genetic cluster",
        ylab = "Fecundity score",
        las = 2,            # vertical labels
        cex.axis = 0.6)     # smaller axis text
```




```
boxplot(larvae_fabae ~ cluster_name, data = merged_cluster,
        main = "Box Plot by Cluster (A. fabae)",
        xlab = "Genetic cluster",
        ylab = "Fecundity score",
        las = 2,            # vertical labels
        cex.axis = 0.6)     # smaller axis text
```




```
anova_model <- aov(larvae_pisum ~ cluster_name, data = merged_cluster)
summary(anova_model)
```

```
              Df Sum Sq Mean Sq F value Pr(>F)
cluster_name   5  234.8   46.96   3.844 0.0024 **
Residuals    198 2418.7   12.22
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
library(multcomp)
```

```
Loading required package: mvtnorm
```

```
Loading required package: survival
```

```
Loading required package: TH.data
```

```
Loading required package: MASS
```

```
Attaching package: 'MASS'
```

```
The following object is masked from 'package:dplyr':

    select

Attaching package: 'TH.data'

The following object is masked from 'package:MASS':

    geyser
```

```r
posthoc <- glht(anova_model, linfct = mcp(cluster_name = "Tukey"))
summary(posthoc)
```

```
     Simultaneous Tests for General Linear Hypotheses

Multiple Comparisons of Means: Tukey Contrasts


Fit: aov(formula = larvae_pisum ~ cluster_name, data = merged_cluster)

Linear Hypotheses:
                               Estimate Std. Error t value Pr(>|t|)
North - East == 0               -1.1426     0.8548  -1.337   0.7428
Paucijuga - East == 0           -4.2564     1.7392  -2.447   0.1305
South - East == 0               -1.8077     0.8773  -2.061   0.2877
South_America - East == 0        1.4012     1.1815   1.186   0.8269
Winter - East == 0               0.3741     1.1598   0.323   0.9995
Paucijuga - North == 0          -3.1138     1.6100  -1.934   0.3574
South - North == 0              -0.6651     0.5804  -1.146   0.8465
South_America - North == 0       2.5437     0.9815   2.592   0.0926 .
Winter - North == 0              1.5167     0.9552   1.588   0.5802
South - Paucijuga == 0           2.4487     1.6220   1.510   0.6325
South_America - Paucijuga == 0   5.6575     1.8048   3.135   0.0210 *
Winter - Paucijuga == 0          4.6305     1.7907   2.586   0.0941 .
South_America - South == 0       3.2089     1.0011   3.205   0.0167 *
Winter - South == 0              2.1818     0.9754   2.237   0.2052
Winter - South_America == 0     -1.0271     1.2561  -0.818   0.9592
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Adjusted p values reported -- single-step method)
```

```r
# Assign letters to groups
library(agricolae)
tukey <- TukeyHSD(anova_model)
tukey
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = larvae_pisum ~ cluster_name, data = merged_cluster)

$cluster_name
                              diff        lwr        upr     p adj
North-East               -1.1425729 -3.6025068  1.3173610 0.7642363
Paucijuga-East           -4.2563865 -9.2614942  0.7487212 0.1453464
South-East               -1.8077055 -4.3323737  0.7169626 0.3124143
South_America-East        1.4011556 -1.9991454  4.8014567 0.8432090
Winter-East               0.3740993 -2.9636416  3.7118403 0.9995304
Paucijuga-North          -3.1138136 -7.7470995  1.5194723 0.3845849
South-North              -0.6651327 -2.3355172  1.0052519 0.8614515
South_America-North       2.5437285 -0.2808569  5.3683139 0.1041218
Winter-North              1.5166722 -1.2322821  4.2656265 0.6075867
South-Paucijuga           2.4486810 -2.2192964  7.1166583 0.6585059
South_America-Paucijuga   5.6575421  0.4634964 10.8515878 0.0239598
Winter-Paucijuga          4.6304858 -0.5228216  9.7837933 0.1055815
South_America-South       3.2088612  0.3277230  6.0899993 0.0193393
Winter-South              2.1818049 -0.6252263  4.9888361 0.2257460
Winter-South_America     -1.0270563 -4.6419591  2.5878465 0.9640196
```

```r
hsd <- HSD.test(anova_model, "cluster_name", group = TRUE)
hsd$groups
```

```
              larvae_pisum groups
South_America     30.06343      a
Winter            29.03637     ab
East              28.66227     ab
North             27.51970     ab
South             26.85456      b
Paucijuga         24.40588      b
```

```r
anova_model <- aov(larvae_pisum ~ cluster_name, data = merged_cluster)
summary(anova_model)
```

```
              Df Sum Sq Mean Sq F value Pr(>F)
cluster_name   5  234.8   46.96   3.844 0.0024 **
Residuals    198 2418.7   12.22
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
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```r
library(multcomp)
posthoc <- glht(anova_model, linfct = mcp(cluster_name = "Tukey"))
summary(posthoc)
```

```
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South_America - East == 0        1.4012     1.1815   1.186   0.8269
Winter - East == 0               0.3741     1.1598   0.323   0.9995
Paucijuga - North == 0          -3.1138     1.6100  -1.934   0.3575
South - North == 0              -0.6651     0.5804  -1.146   0.8466
South_America - North == 0       2.5437     0.9815   2.592   0.0927 .
Winter - North == 0              1.5167     0.9552   1.588   0.5803
South - Paucijuga == 0           2.4487     1.6220   1.510   0.6325
South_America - Paucijuga == 0   5.6575     1.8048   3.135   0.0208 *
Winter - Paucijuga == 0          4.6305     1.7907   2.586   0.0939 .
South_America - South == 0       3.2089     1.0011   3.205   0.0170 *
Winter - South == 0              2.1818     0.9754   2.237   0.2052
Winter - South_America == 0     -1.0271     1.2561  -0.818   0.9592
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
(Adjusted p values reported -- single-step method)
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library(agricolae)
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  Tukey multiple comparisons of means
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Fit: aov(formula = larvae_pisum ~ cluster_name, data = merged_cluster)

$cluster_name
                              diff        lwr        upr     p adj
North-East               -1.1425729 -3.6025068  1.3173610 0.7642363
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South-North              -0.6651327 -2.3355172  1.0052519 0.8614515
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Winter-North              1.5166722 -1.2322821  4.2656265 0.6075867
South-Paucijuga           2.4486810 -2.2192964  7.1166583 0.6585059
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```

```r
hsd <- HSD.test(anova_model, "cluster_name", group = TRUE)
hsd$groups
```

```
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South_America     30.06343      a
Winter            29.03637     ab
East              28.66227     ab
North             27.51970     ab
South             26.85456      b
Paucijuga         24.40588      b
```

4 The script used for fecundity figure (Figure 1)

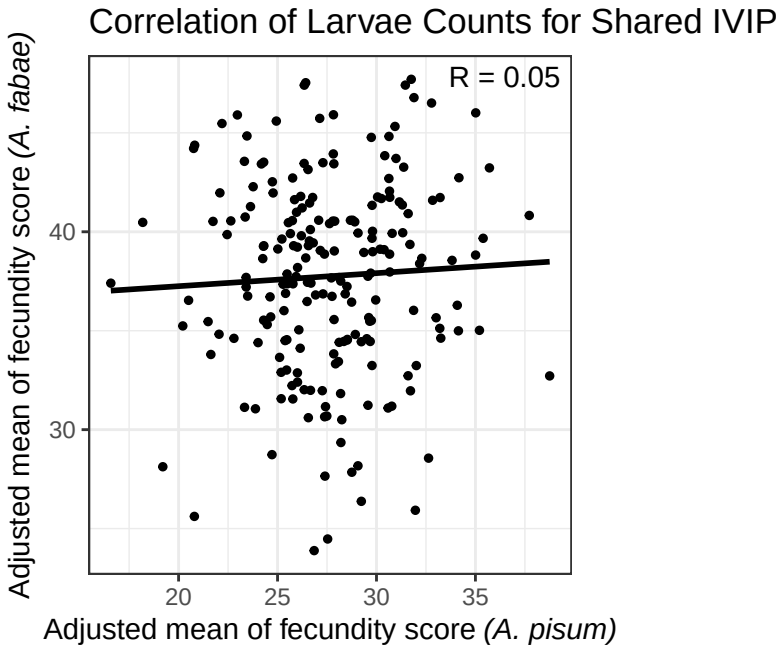
```
cor_test <- cor.test(merged$larvae_pisum, merged$larvae_fabae)
cor_test
```

```
    Pearson's product-moment correlation

data:  merged$larvae_pisum and merged$larvae_fabae
t = 0.69667, df = 204, p-value = 0.4868
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.08857225  0.18419352
sample estimates:
       cor
0.04871894
```

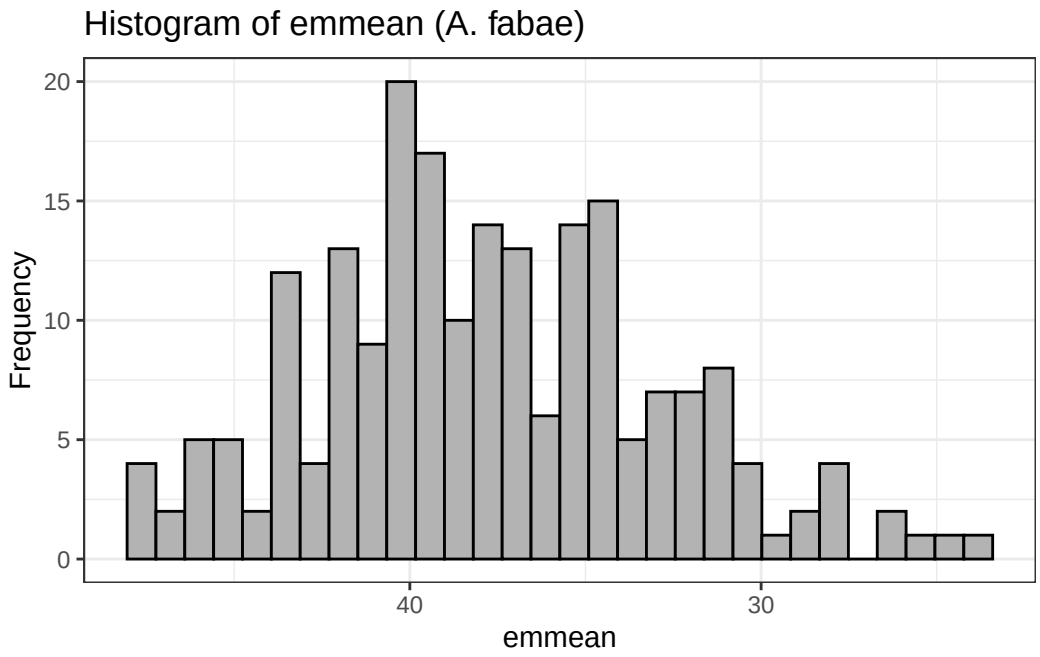
```
ggplot(merged, aes(x = larvae_pisum, y = larvae_fabae)) +
  geom_point(size = 1) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  annotate(
    "text",
    x = Inf, y = Inf,
    label = paste0("R = ", round(cor_test$estimate, 2)),
    hjust = 1.1, vjust = 1.5
  ) +
  labs(
    title = "Correlation of Larvae Counts for Shared IVIP",
    x = expression("Adjusted mean of fecundity score" ~ italic("(A. pisum)")),
    y = expression("Adjusted mean of fecundity score" ~ italic("(A. fabae)"))
  ) +
  coord_fixed(ratio = 1) +
  theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

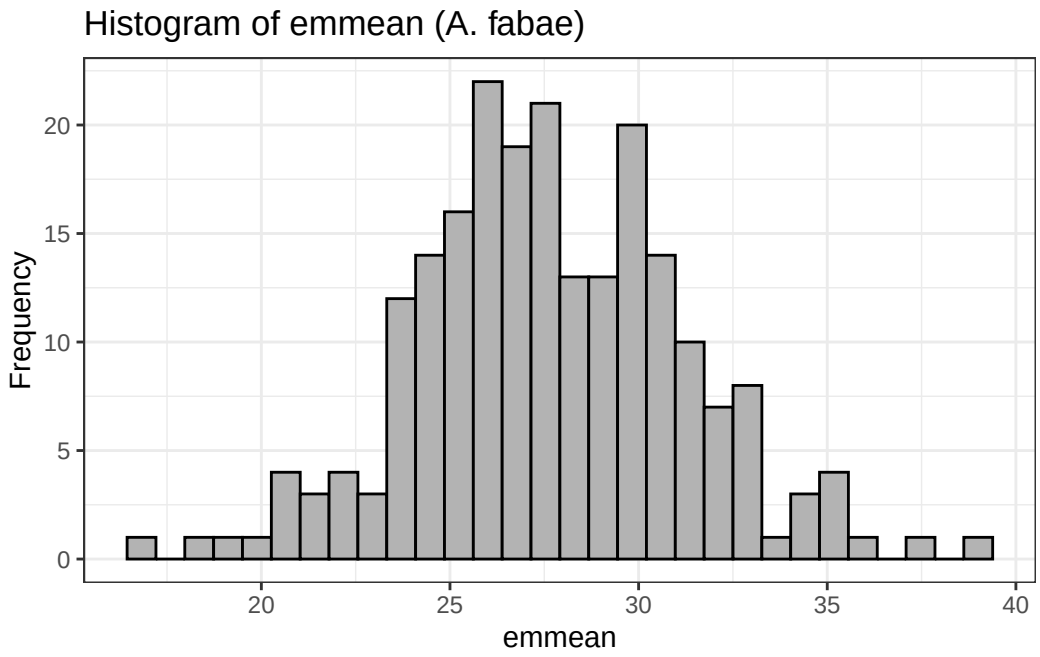



```
fabae<-read.csv("../../../01.GWAS/01_phenotipic_analysis/02.ajusted_means/fabae/emmeans_faba
pisum <- read.csv("../../../01.GWAS/01_phenotipic_analysis/02.ajusted_means/pisum/emmeans_pi

ggplot(fabae, aes(x = emmean)) +
  geom_histogram(
    bins = 30,
    color = "black",
    fill = "grey70"
  ) +
  scale_x_reverse() +
  labs(
    title = "Histogram of emmean (A. fabae)",
    x = "emmean",
    y = "Frequency"
  ) +
  theme_bw()
```




```
ggplot(pisum, aes(x = emmean)) +
  geom_histogram(
    bins = 30,
    color = "black",
    fill = "grey70"
  ) +
  labs(
    title = "Histogram of emmean (A. fabae)",
    x = "emmean",
    y = "Frequency"
  ) +
  theme_bw()
```


Histogram of emmean (A. fabae)
Frequency
0
5
10
15
20
20
25
30
35
40
emmean